

CPS 150 Review Session #5

Loops

Review Activity

What will this loop output?

```
String word = "Hello123";  
int index = 0;  
  
while (Character.isLetter(word.charAt(index))) {  
    System.out.println(word.charAt(index));  
    index++;  
}
```

For Loops

For loops allow us to quickly index or iterate through something.

```
for (statement_1; statement_2; statement_3) {  
    // code to be ran  
}
```

While loop equivalent

```
statement_1;  
while (statement_2) {  
    // code to be ran  
    statement_3;  
}
```

For Loops

Fill in the missing pieces to get the desired output

```
for (int i = _; i _ _; i____) {  
    System.out.print(i + " ");  
}
```

Desired Outputs:

1 3 5 7 9

10 9 8 7 6 5 4 3 2 1

1 2 4 8 16 32 64

For Loops

What happens when we run this code?

```
for (int i = 0; i < 5; i++) {  
    System.out.println(i + " ");  
}  
  
System.out.println("Final Value: " + i);
```

Programming Practice

Write a program that prints out a reversed version of a user input string

Example:

```
Enter a string: Java  
avaJ
```

Programming Practice

Write a program that counts the number of uppercase and lowercase letters in a user inputted string

Example:

```
Enter a string: Hello World!  
The string contains 2 uppercase letters and 8 lowercase letters
```